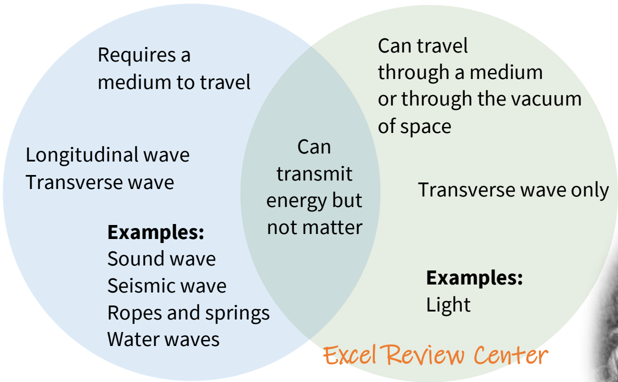
Physics Part 4

Mechanical Wave / Electromagnetic Wave



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"Every point on a wave front may be considered as a new source of disturbance, sending wavelets in forward directions."

This is known as Huygen's Principle.

Named after the Dutch mathematician and physicist, Christiaan Huygens

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Reflection and Refraction

Reflection is the bouncing off of a wave as it hits a barrier. The angle that the wave hits the barrier should be the equal to the angle at which it bounces off the barrier. That is the most basic premise of the **Law of Reflection**.

Refraction is the changing in the direction of a wave as it travels from one medium to a different medium. Every medium or material has a property known as the **index of refraction** which is the ratio of the speed of light in free space to the velocity of the wave in the actual medium.

$$n = \frac{c}{v}$$

What is Frequency

Frequency (f) is the number of waves per unit time that pass a point. **Period** (T) is the time required for one wave to pass a point. Just like simple harmonic motion, the period is reciprocal with the frequency and vice versa.

$$f = \frac{1}{T}$$

Wavelength is the distance between one particle in a wave and the corresponding particle in the next wave. Two particles are in the same phase if they have the same displacement and are moving in the same direction. Relation between speed v, frequency f and wavelength λ:

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$$v = f\lambda$$

Intensity of wave refers to the energy transferred per unit time per unit area through a surface perpendicular to the direction of the motion of the wave.

$$I = 2\pi^2 v \rho f^2 A^2$$

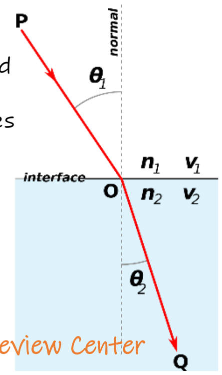
where:

- v - speed of wave (m/s)
- ρ - density (kg/m³)
- f – frequency (Hz)
- A – amplitude (m)

Snell's Law

Named after the Dutch astronomer and mathematician, Willebrord Snellius.

Sometimes known as the **Law of Refraction**, it states that the ratio of the sines of the angles of incidence and the angle of refraction is a constant that depends on the ratio of the indices of both media.



$$n_1 \sin\theta_i = n_2 \sin\theta_r$$

where:

- n_1 – refractive index of medium 1
- n_2 – refractive index of medium 2
- θ_i - angle of incidence = θ_1
- θ_r - angle of refraction = θ_2

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It is a mechanical wave that carries disturbances from one location to another through a certain medium. Most of the time, this medium is air, though it could be any other media such as water or steel. Sound is also classified as a longitudinal wave since the particles of the medium through which the sound moves vibrate parallel to the direction of the sound.

Frequency and Pitch of Sound

For sound, the particles of the medium that moves back and forth will determine its frequency. 1 Hertz is 1 vibration per second. The human ear is capable of detecting sound waves with a very wide range of frequencies, ranging from 20 Hz to 20 kHz. Sound with frequencies below 20 Hz is called infrasound while sound with frequencies beyond 20 kHz is called ultrasound. These frequencies are now outside the audible range for the human ear. Pitch is synonymous with frequency.

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Intensity of Sound

The **intensity of sound** refers to the amount of energy that is transferred per unit are and per unit time. The SI unit of intensity is W/m².

$$\text{Intensity} = \frac{\text{Energy}}{\text{Area} \times \text{Time}} = \frac{\text{Power}}{\text{Time}}$$

The lowest intensity that the human ear can detect is around $1\text{x}10^{-12}$ W/m². This is also known as the **threshold of hearing**. A more practical way to measure intensity is to do it in the decibel scale, with $1\text{x}10^{-12}$ W/m² being 0 dB.

$$\text{Intensity (dB)} = 10\log\frac{\text{Intensity}}{1\times10^{-12}\,\frac{W}{m^2}}$$

Speed of Sound

The speed of the sound wave is the rate at which the disturbance in the medium is carried on from one particle to the next. The faster the sound travels, the more distance it covers in a certain period of time. There are different factors that can affect the speed of sound. First, the properties of the medium in which the sound travels greatly affects the speed of sound. The phase of the medium affects the speed as well. It is established that sound travels fastest in solids and slowest in gases. Aside from that, if the medium is air, the temperature and even the humidity can also affect the speed of sound. With E = elastic constant of medium, ρ = density of medium or gas, T = temperature of gas, P = pressure of gas, R = universal gas constant, M = molecular mass of gas.

Speed of sound if the medium is a thin rod: $v = \sqrt{\frac{E}{\rho}}$

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Speed of sound if the medium is a gas: $v = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{M}}$

At standard temperature of 0°C, the speed of sound in air is $v \approx 331\text{ m/s} \approx 1{,}100\text{ ft/s}$

At any other temperature: $v = 331 + 0.6T$

Where: T - temperature of air in degrees Celsius